

CS 182 / EECS 282A Official Course Notes

Fall 2026

Note 2: Least-Squares Optimization, Ridge, and Implicit Regularization

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Prerequisites

This note assumes matrix calculus for quadratic objectives, eigenvalues and the singular value decomposition, and orthogonal projections, row spaces, and null spaces. For a concise review, consult the JRS pages listed below.

Reading and coverage map

Prince correspondence (direct: Section 6.1 and Sections 9.1–9.3). Prince covers gradient-based optimization and explicit/implicit regularization. This note makes the least-squares SVD dynamics and minimum-norm selection fully explicit.

JRS prerequisite review (pp. 55–56, 71–73, 126–127, and 134–135). These pages review least squares, the pseudoinverse/SVD, ridge regression, and regularization concepts used in the derivations.

Beyond Prince. This note develops the paired spectral-filter formulas for ridge and early stopping, the row-space/null-space bookkeeping, and the minimum-norm derivation in detail.

2.1 Least squares as a dynamical system

Least squares is one of our lampposts: a tractable place to look because its geometry is explicit and linear algebra lets us see the dynamics. The discipline is tracking which conclusions belong to the quadratic model and which reasoning moves travel to nonlinear networks.

Let $X \in \mathbb{R}^{n \times d}$, $\vec{y} \in \mathbb{R}^n$, and $\vec{w} \in \mathbb{R}^d$. For the unnormalized squared loss

$$L(\vec{w}) = \|X\vec{w} - \vec{y}\|_2^2, \quad \nabla L(\vec{w}) = 2X^\top(X\vec{w} - \vec{y}), \quad (2.1)$$

gradient descent with learning rate η gives

$$\vec{w}_{t+1} = (I - 2\eta X^\top X)\vec{w}_t + 2\eta X^\top \vec{y}. \quad (2.2)$$

Let $\vec{w}_{\min} = X^\dagger \vec{y}$ denote the minimum-Euclidean-norm least-squares solution. It satisfies

$$X^\top X \vec{w}_{\min} = X^\top \vec{y}, \quad \vec{w}_{\min} \in \text{Row}(X). \quad (2.3)$$

For $\Delta_t = \vec{w}_t - \vec{w}_{\min}$,

$$\Delta_{t+1} = (I - 2\eta X^\top X)\Delta_t. \quad (2.4)$$

This is the central simplification: optimization of a quadratic becomes a linear dynamical system.

In the classical $n > d$ regime, full column rank makes the least-squares coefficient unique. In the overparameterized $d \gg n$ regime, even full row rank leaves an affine family of interpolating parameters. The update span and initialization then select among many solutions. This is the source of the implicit-bias question developed below and summarized geometrically in Figure ??.

2.1.1 Contraction occurs only in the row space

To understand which parts of $\vec{w}$ gradient descent can actually change, we need two subspaces of $\mathbb{R}^d$:

- The **row space** $\text{Row}(X)$: the span of the rows of X . Equivalently, $\text{Row}(X) = \text{Range}(X^\top)$, the set of all vectors that can be written as $X^\top \vec{v}$ for some $\vec{v} \in \mathbb{R}^n$.
- The **null space** $\text{Null}(X)$: the set of all vectors $\vec{z} \in \mathbb{R}^d$ satisfying $X\vec{z} = \vec{0}$. Moving $\vec{w}$ in a null-space direction does not change the predictions $X\vec{w}$ at all.

These two subspaces are orthogonal complements: every $\vec{w} \in \mathbb{R}^d$ splits uniquely as

$$\vec{w} = \underbrace{P_{\text{Row}(X)}\vec{w}}_{\text{row-space component}} + \underbrace{P_{\text{Null}(X)}\vec{w}}_{\text{null-space component}},$$

where $P_{\text{Row}(X)}$ and $P_{\text{Null}(X)}$ are the orthogonal projection matrices onto the respective subspaces.

Now look at the gradient descent update from Equation (2.2):

$$\vec{w}_{t+1} = \vec{w}_t - 2\eta \underbrace{X^\top (X\vec{w}_t - \vec{y})}_{\text{this vector is } X^\top(\text{something})}.$$

The update term $X^\top (X\vec{w}_t - \vec{y})$ is of the form $X^\top \vec{v}$, which by definition lies in $\text{Range}(X^\top) = \text{Row}(X)$. So gradient descent can only push $\vec{w}$ along directions in the row space of X . The null-space component of $\vec{w}$ is never touched:

$$P_{\text{Null}(X)}\vec{w}_t = P_{\text{Null}(X)}\vec{w}_0 \quad \text{for every } t. \quad (2.5)$$

In words: *whatever null-space component the initialization has, gradient descent preserves it forever.* This is why initializing at $\vec{w}_0 = \vec{0}$ matters in the overparameterized regime—it forces the null-space component to be zero for all time, steering gradient descent toward the minimum-norm solution $\vec{w}_{\min} = X^\dagger \vec{y}$. (See Figure 2.1.)

Let $0 < \lambda_1 \leq \dots \leq \lambda_r = \lambda_{\max}$ be the positive eigenvalues of $X^\top X$. Each corresponding error coordinate is multiplied by $1 - 2\eta\lambda_i$, so

$$\rho(\eta) = \max_{1 \leq i \leq r} |1 - 2\eta\lambda_i|, \quad \|P_{\text{Row}(X)}\Delta_t\|_2 \leq \rho(\eta)^t \|P_{\text{Row}(X)}\Delta_0\|_2. \quad (2.6)$$

The exact constant-step condition is

$$0 < \eta < \frac{1}{\lambda_{\max}(X^\top X)} = \frac{1}{\sigma_{\max}(X)^2}. \quad (2.7)$$

For this loss convention, the best constant learning rate on the positive-eigenvalue subspace is chosen so the two extreme modes contract equally with opposite-signed factors,

$$1 - 2\eta_*\lambda_{\min}^+ = -(1 - 2\eta_*\lambda_{\max}). \quad (2.8)$$

The slowest mode is still shrinking while the fastest just overshoots. Solving and substituting gives

$$\eta_* = \frac{1}{\lambda_{\max} + \lambda_{\min}^+}, \quad \rho(\eta_*) = \frac{\kappa - 1}{\kappa + 1}, \quad \kappa = \frac{\lambda_{\max}}{\lambda_{\min}^+}. \quad (2.9)$$

Large condition number therefore means slow convergence even when the iteration is stable.

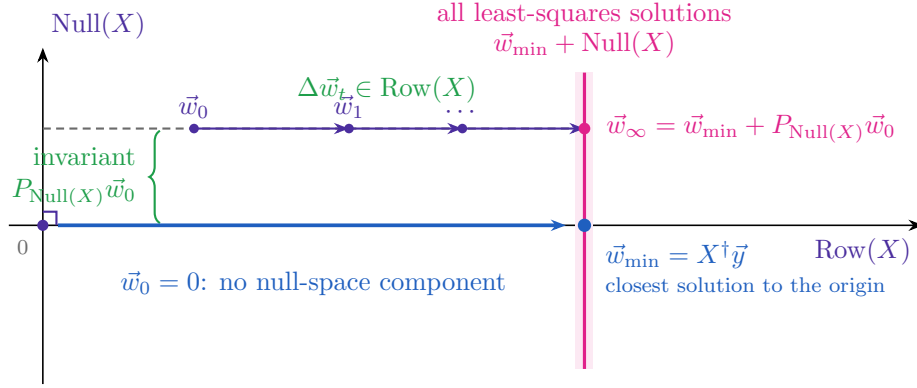


Figure 2.1: Gradient descent can move only along $\text{Row}(X)$, so the null-space coordinate of the initialization is invariant. The affine family of least-squares solutions is $\vec{w}_{\min} + \text{Null}(X)$; zero initialization follows the row space to its minimum-norm member $\vec{w}_{\min}$.

Remark

Starting at $\vec{w}_0 = 0$ removes the null-space component, so gradient descent converges to $\vec{w}_{\min}$. From a general initialization it converges to $\vec{w}_{\min} + P_{\text{Null}(X)} \vec{w}_0$.

2.2 Ridge regression in SVD coordinates

Ridge regression adds an explicit quadratic penalty:

$$\vec{w}_{\text{ridge}} = \arg \min_{\vec{w}} (\|X\vec{w} - \vec{y}\|_2^2 + \lambda \|\vec{w}\|_2^2), \quad \lambda > 0. \quad (2.10)$$

Its primal and kernel forms are

$$\vec{w}_{\text{ridge}} = (X^\top X + \lambda I_d)^{-1} X^\top \vec{y} = X^\top (X X^\top + \lambda I_n)^{-1} \vec{y}. \quad (2.11)$$

From $X^\top (X X^\top + \lambda I_n) = (X^\top X + \lambda I_d) X^\top$, multiplying by the two inverses yields the second form.

Take a full SVD $X = U \Sigma V^\top$, with singular values $\sigma_1, \dots, \sigma_{\min(n,d)}$. Substitution in Equation (2.11) gives

$$\vec{w}_{\text{ridge}} = \sum_{i=1}^{\min(n,d)} v_i \frac{\sigma_i}{\sigma_i^2 + \lambda} u_i^\top \vec{y}. \quad (2.12)$$

The pseudoinverse coefficient is $(u_i^\top \vec{y})/\sigma_i$. Relative to it, ridge retains the fraction

$$r_\lambda(\sigma_i) = \frac{\sigma_i^2}{\sigma_i^2 + \lambda}. \quad (2.13)$$

Directions with $\sigma_i^2 \gg \lambda$ are nearly preserved, while directions with $\sigma_i^2 \ll \lambda$ are strongly attenuated.

The data have high energy along large-singular-value directions and low energy along small-singular-value directions. Suppressing the latter can improve generalization when noise dominates those poorly constrained directions. Small-singular-value directions can still contain real signal, so deciding to suppress them is a modeling judgment.

Remark

Overfitting is getting “too married” to the training data: loving the observed sample so much that one uses weakly represented directions to fit the observed sample perfectly when variation along those directions is dominated by noise. Regularization expresses calibrated distrust of those poorly constrained directions, while leaving open the possibility that some contain real signal.

A probabilistic model supplies one route to the same penalty. Suppose

$$\vec{y} \mid X, \vec{w} \sim \mathcal{N}(X\vec{w}, \sigma_y^2 I_n), \quad \vec{w} \sim \mathcal{N}(0, \tau^2 I_d). \quad (2.14)$$

Up to constants independent of $\vec{w}$, the negative log joint density is

$$-\log p(\vec{y}, \vec{w} \mid X) = \frac{1}{2\sigma_y^2} \|X\vec{w} - \vec{y}\|_2^2 + \frac{1}{2\tau^2} \|\vec{w}\|_2^2 + \text{constant}. \quad (2.15)$$

Its $\vec{w}$ -minimizer is the MAP estimate, so MAP estimation minimizes the unnormalized ridge objective with $\lambda = \sigma_y^2/\tau^2$. If the data loss is averaged over n examples, the corresponding coefficient is $\sigma_y^2/(n\tau^2)$. Figure ?? places this MAP route beside the earlier interpolation geometry.

A bare nonnegative penalty multiplier is not learned by minimizing $\|X\vec{w} - \vec{y}\|_2^2 + \lambda\|\vec{w}\|_2^2$ jointly over $\vec{w}$ and λ . At fixed $\vec{w}$ that objective is nondecreasing in λ , so joint minimization collapses to the boundary $\lambda = 0$ unless the weight norm is already zero. Select λ with held-out data, or specify a genuine hierarchical probabilistic model whose variance parameters bring their own normalization terms.

2.2.1 Ridge gradient descent is weight decay

Applying gradient descent directly to the ridge objective yields

$$\vec{w}_{t+1} = (1 - 2\eta\lambda)\vec{w}_t - 2\eta X^\top (X\vec{w}_t - \vec{y}). \quad (2.16)$$

The first factor shrinks the current parameter at every step. For plain gradient descent on this quadratic, ridge regularization and this coupled weight-decay update are two views of the same calculation.

2.3 Early stopping as implicit regularization

Write $\tilde{w}_t = V^\top \vec{w}_t$ and $\tilde{y} = U^\top \vec{y}$, so we are expressing gradient descent in the SVD basis. Substituting the full SVD $X = U\Sigma V^\top$ into Equation (2.2) and multiplying by $V^\top$ on the left yields

$$\begin{aligned} V^\top \vec{w}_{t+1} &= V^\top (I - 2\eta V \Sigma^\top \Sigma V^\top) \vec{w}_t + 2\eta V^\top V \Sigma^\top U^\top \vec{y} \\ &= (I - 2\eta \Sigma^\top \Sigma) V^\top \vec{w}_t + 2\eta \Sigma^\top U^\top \vec{y} \\ \tilde{w}_{t+1} &= (I - 2\eta \Sigma^\top \Sigma) \tilde{w}_t + 2\eta \Sigma^\top \tilde{y}. \end{aligned}$$

This diagonalized vector equation decouples into scalar recurrences

$$\tilde{w}_{t+1,i} = (1 - 2\eta\sigma_i^2) \tilde{w}_{t,i} + 2\eta\sigma_i \tilde{y}_i. \quad (2.17)$$

With $\vec{w}_0 = 0$, the exact solution is

$$\tilde{w}_{t,i} = \frac{1 - (1 - 2\eta\sigma_i^2)^t}{\sigma_i} \tilde{y}_i. \quad (2.18)$$

The scalar multiplier is

$$q_t(\sigma_i) = 1 - (1 - 2\eta\sigma_i^2)^t. \quad (2.19)$$

With an appropriately chosen η : large singular values converge first and small singular values move more slowly.¹ In that regime, stopping before convergence therefore has the same qualitative effect as ridge: fit strongly observed directions while attenuating weakly observed ones (compare the filter shapes in Figure 2.2).

Remark

Singular values commonly span orders of magnitude. When diagnosing conditioning, a linear-scale plot can make the small end appear identically zero or hide the separation that explains slow directions. Plotting singular values on a logarithmic scale is therefore a useful debugging habit. Remember that exact zeros matter for rank.

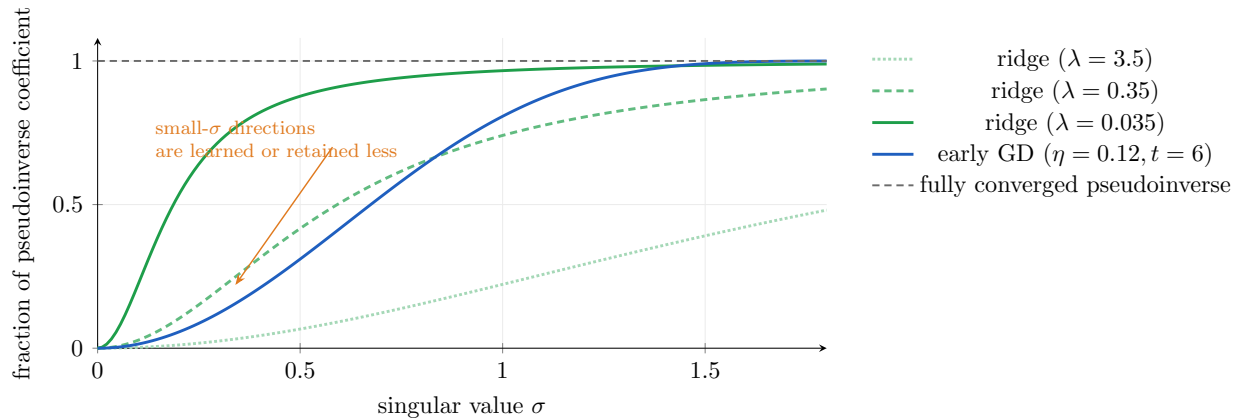


Figure 2.2: Ridge and early stopping are different spectral filters. The curves use illustrative values of λ , η , and t . The comparison is between the formulas, and the curves need not coincide pointwise. Both suppress small-singular-value directions relative to the fully converged pseudoinverse.

Pitfall

Ridge and early stopping are not literally the same estimator. Their filter shapes differ, and early stopping also depends on the initialization and learning-rate convention. Their shared lesson is directional: regularization changes how rapidly or strongly different singular directions enter the solution.

¹For all singular directions, this monotone interpretation is guaranteed by $0 < \eta \leq 1/(2\sigma_{\max}^2)$. In the broader stable range $1/(2\sigma_{\max}^2) < \eta < 1/\sigma_{\max}^2$, large-singular-value coordinates can oscillate or overshoot, so the ordering need not remain monotone.

2.4 Selecting the control knobs

Regularization strength, learning rate, and stopping time are hyperparameters. Their useful range often spans orders of magnitude, so logarithmic sweeps are more informative than linear ones. A validation set selects among trained candidates; the test set is reserved for the resulting fixed procedure. Exhaustive grids scale poorly with the number of hyperparameters, making random or adaptive search attractive. When broad search is unaffordable, coupled scaling laws and settings from closely related models become engineering priors rather than mathematical guarantees.

The pragmatic advice is to be superstitious: when broad search is unaffordable, borrow settings from a closely related paper or implementation as an engineering prior. Superstition supplies a starting heuristic, neither a derivation nor evidence that the settings are optimal for the new system.

Remark

Can another learner choose or vary the hyperparameters? Meta-learning and adaptive search can do this, but they remain outer procedures with their own training data, validation design, and computational cost. They improve the search strategy; they do not eliminate the search problem.

In practice, early stopping usually means checkpointing throughout training and selecting the checkpoint with the best validation performance. It does not mean choosing an arbitrary small iteration count without looking at generalization.